

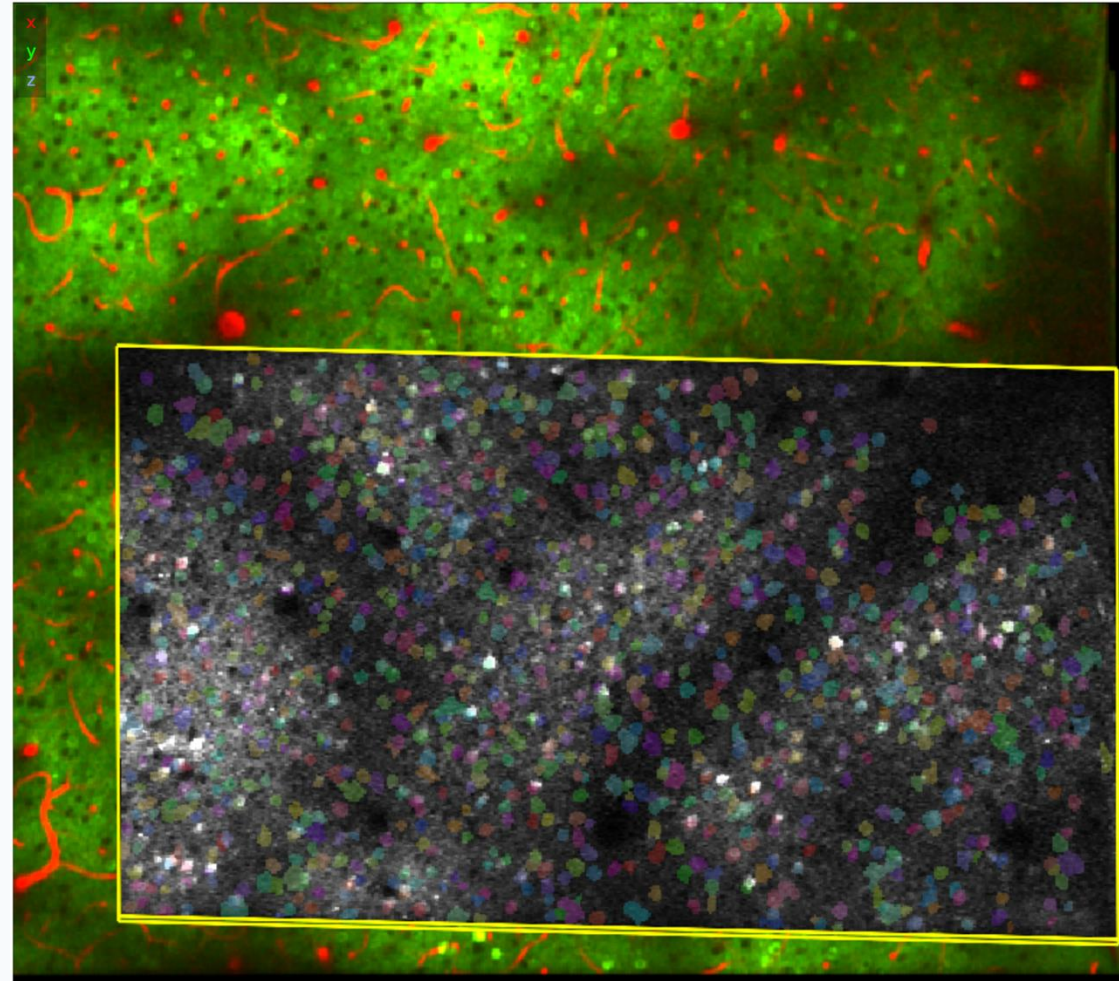
Automated Neuronal ROI Segmentation in MICrONS Two-Photon Imaging

Midterm Blueprint · ITAI 1378 Computer Vision-Artificial Intel

**Two-photon movie frames → segmentation
model → predicted neuronal ROI masks**

Tier 1 · single-model semantic segmentation application

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[Image source: MICrONS Explorer, Functional Data](#)

The Problem

Why automated neuronal-region segmentation matters?

Real-world problem

Two-photon microscopy can produce many images or frames. Manually identifying neuronal regions is repetitive, time-consuming, and can vary between observers.

Who cares?

Neuroscience researchers who need consistent regions of interest before extracting and analyzing calcium activity.

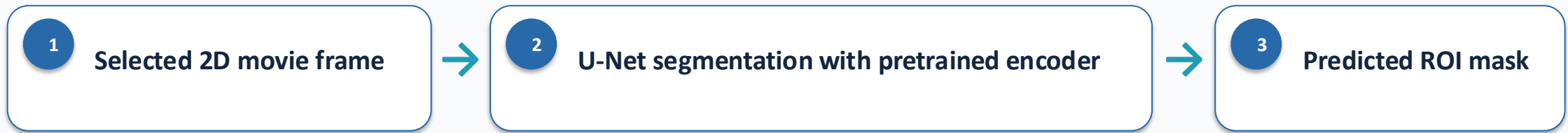
What will solve it?

A computer-vision pipeline that accepts a processed two-photon frame and automatically predicts neuronal ROI masks for visualization and quantitative evaluation.

Solution

A small end-to-end computer vision application

APPLICATION PIPELINE



EVALUATION



Final output

Original frames | Predicted mask | Reference Mask | Visual Comparison | IoU / Dice

The application takes a processed MICrONS two-photon image as input, generates a predicted neuronal ROI mask, and compares the prediction with the MICrONS reference segmentation using IoU and Dice scores.

Technical Approach

CV technique

Semantic segmentation — predict a class label for each pixel to separate neuronal regions from background.

Input
2D grayscale two-photon frame

Output
binary neuronal ROI mask for the initial baseline

Model

U-Net segmentation in PyTorch.

Use an established U-Net implementation with a pretrained encoder if compatible with the microscopy input

Fine-tune on the selected MICrONS subset.

Why this approach

ROI boundaries are pixel-level structures.

The desired output is a spatial mask, not only an image-level bounding box.

U-Net is designed for biomedical image segmentation and works with relatively limited labeled data.

Preprocess selected microscopy images into a consistent input format.

Run inference to produce a binary or labeled neuronal-region mask.

Compare predicted masks with available reference masks using overlap-based metrics.

Visualize original image, reference mask, prediction, and overlay.

Data Plan

MICrONS source

- 19 functional two-photon scans
- GCaMP6s excitatory-neuron imaging
- raster + motion corrected TIFFs
- processed cell segmentation masks available
- 66–95 GB per scan
- ~1.3 TB total TIFF imaging data

Project subset

- choose one scan / imaging field
- extract a manageable set of 2D frames or summary images
- retrieve the published ROI segmentation for the same imaging field
- prepare aligned image-mask pairs for training/evaluation
- keep only data needed for training/evaluation

USE movie frames + published segmentation masks

NOT USE full movies, visual stimuli, calcium traces, pupil traces, behavioral traces, EM reconstruction

Data source:

MICrONS Explorer v117 / MICrONS Tutorial Static Repositories — raster- and motion-corrected two-photon TIF scans + processed cell segmentation masks.

<https://www.microns-explorer.org/cortical-mm3>

Success Metrics

Measure overlap between predicted and published masks

Primary: Intersection over Union (IoU)

$\text{IoU} = \text{overlap} / \text{union}$

Initial target: mean IoU ≥ 0.60

Why: IoU provides a strict measure of segmentation agreement by comparing the shared region with the total area covered by either mask.

Secondary: Dice coefficient

$\text{Dice} = 2 \times \text{overlap} / \text{total mask area}$

Initial target: mean Dice ≥ 0.70

Why: Dice provides a complementary overlap measure that emphasizes similarity between the predicted and reference masks.

Qualitative check

Frame | Reference Mask | Predicted Mask | Overlay

Why both methods? Metrics quantify overlap; overlays show where the model misses or over-segments neuronal regions.

Milestone Plan

10-week term structure

Week 5	Blueprint	Proposal, GitHub repo, data source and metrics defined
Week 6	First Working Demo	Run an end-to-end segmentation baseline on 5-10 sample images
Weeks 7–8	Pipeline	Pair the selected MICrONS subset frames with masks; refine preprocessing, train or fine-tune if needed
Week 9	Improve & Measure	Compute IoU/Dice; inspect failure cases; make focused improvements
Week 10	Build & Present	Finalized notebook, Demo, GitHub, README, final slides

Key principle: the first working pipeline comes before heavy data preparation or training.

8. Risks, Resources, and Future Extension

Risk 1 · Data access / format

MICrONS files are large, and the functional data format may require additional preprocessing.

Plan B: use a small exported set of representative images and matching ROI masks rather than downloading full scans.

Risk 2 · Model / data compatibility

A pretrained segmentation model may not transfer well to two-photon microscopy images.

Plan B: train a small U-Net baseline on the selected subset and reduce the project to a one-field proof of concept.

Resources

Python · PyTorch · NumPy · OpenCV · Matplotlib · Google Colab · GitHub

Compute: Google Colab GPU when available

Estimated cost: \$0

Future Extension:

ROI segmentation → extract ROI calcium traces → visualize neuronal activity over time

Possible capstone: integrate segmentation with calcium-trace analysis in the same imaging field.